

1. DEFINITION

DEFINITION	
Glycerol mono-oleate (Type 40) EP / Glyceryl monooleate (Type 40) NF/ChP	

DENOMINATION	
INCI NAME* (PCPC - International Nomenclature of Cosmetic Ingredients)	Glyceryl Oleate
UNII (Unique Ingredient Identifier) USP/FDA Substance Registration System (SRS)	4PC054V79P (Glyceryl Oleate)
*INCI name is given for information only. This ingredient is a pharmaceutical excipient. It must be used according to applicable regulations. Gattefossé accepts no responsibility for the use of this product in applications other than those recommended.	

2. FIELD OF USE

FIELD OF USE
This ingredient must be used according to appropriate regulations
Pharmaceutical ingredient (oral, topical and/or rectal/vaginal routes)

3. MANUFACTURING PROCESS AND COMPOSITION

MANUFACTURING PROCESS
Esterification reaction between Glycerol and Oleic acid. No solvent or catalyst used during the manufacturing process.

COMPOSITION	
Well defined multi-constituent substance composed of mono-, di- and triglycerides of Oleic acid (C18:1), the monoester fraction being predominant (Specification: 32-52%)	
Full Labeling (Key letters system)	
$A_1 \geq 75.0$; $50.0 \leq A_2 < 75.0$; $25.0 \leq B < 50.0$; $10.0 \leq C < 25.0$; $5.0 \leq D < 10.0$; $1.0 \leq E < 5.0$; $0.1 \leq F < 1.0$; $G < 0.1$	
Glyceryl Oleate	$A_1 \geq 75.0\%$
Other constituents (i.e. incidental ingredients)	
None	



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4. DRUG MASTER FILE

Country	Master file number
US DMF (Type IV – Excipient)	22516
Canada MF (Type III – Excipient)	/
Other countries	/
For submission of the DMF in another country, please contact your local Gattefossé representative	

CHINA BUNDLING REVIEW			
(Generic name) Chinese name	Registration N°	Company	CFDA/CDE Reference
(Glyceryl monooleate 40) 单油酸甘油酯	F20200000253	Gattefossé SAS	Raw material has not been approved yet for use in marketed formulations in Joint-Review with drug applications. However, the excipient dossier has already been submitted (Status I).
Please contact our Regulatory Affairs department for further information and obtain letter of authorization			

5. PHARMACOPEIAS

CONFORMITY TO PHARMACOPEIAS	
Reference	Conforms to monograph
European Pharmacopeia Current edition	Glycerol Mono-oleates (Type 40)
USP/ NF Current edition	Glyceryl Monooleate (Type 40)
ChP Current edition	Glyceryl Monooleate (Type 40)



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6. VETERINARY MEDICINES FOR FOOD PRODUCING ANIMALS

EUROPEAN REGULATION Substances authorized to be used in		USA REGULATION Substances authorized to be used in			
Veterinary medicines and/or Biocidal products		Veterinary drugs		Pesticide products	
All products (except *)	* Products for use in food producing animals (husbandry)	All products (except *)	* Drugs for food producing animals (husbandry)	Applied to non-food use sites (e.g., flea and tick products in pets)	Applied to food use sites
✓		✓(5)	✓(5) 21CFR §184.1505	✓(8)	✓(8) 40CFR § 180.930
EUROPEAN REGULATION (1) Regulation (EU) No 37/2010 - Listed in annex II (Food additives: substances with a valid E number approved as additives in food stuffs for human consumption) (2) Regulation (EU) No 37/2010 - Listed in annex II (Polyethylene glycol stearates with 8-40 oxyethylene units) (3) Regulation (EU) No 37/2010 - Listed in annex II (Diethylene glycol monoethyl ether) (4) EMA/CVMP - Substances considered as not falling within the scope of Regulation (EC) No 470/2009, with regard to residues of veterinary medicinal products in foodstuffs of animal origin USA REGULATION - Veterinary drugs FDA is responsible for assuring that animal drugs are not only safe and effective for animals, but that food products from treated animals are safe for humans to consume. In the case of a drug for food producing animals, the inactive ingredients of the drug or its composition may be safe with respect to human food safety (5) Approved food additives and GRAS substances for human food - Title 21, Part 172, 182—184 of the Code of Federal Regulations (CFR) (6) Approved excipients (inactive ingredients) for use in human drugs by oral route (Ref: FDA Inactive ingredient database) - Pesticides (8) EPA's pesticides program (http://iaspub.epa.gov/apex/pesticides/f?p=INERTFINDER:1:0::NO:1::)					

7. SPECIFIC PHARMACEUTICAL REGULATIONS

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS AUSTRALIA (TGA) - Australian Register of Therapeutic Goods Ingredients			
Status	Ingredient Name	Use	Restrictions
Approved	Glyceryl monooleate	Glyceryl Oleate: Available for use as an Active Ingredient in: Biologicals, Prescription Medicines Available for use as an Excipient Ingredient in: Biologicals, Devices, Export Only, Listed Medicines, Over the Counter, Prescription Medicines	/
AUSTRALIA: Therapeutic goods are divided into 'medicines' and 'medical devices'. Some 'medicines' are limited to prescription-only while others are available without a prescription. Non-prescription medicines may be 'complementary medicines' (i.e. vitamin, mineral, herbal, aromatherapy, and homeopathic products) or 'OTC medicines' and may be 'listed' (i.e. Sunscreens SPF4 and >4) or 'registered' in the ARTG.			

REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA NHPID		
Status	NHPID name	Route of administration/ Restrictions
Approved chemical name	Glyceryl oleate	Roles Non-medicinal: Purposes: Emulsifying Agent , Flavour Enhancer , Fragrance Ingredient , Humectant , Skin-Conditioning Agent - Emollient , Solvent Detail: "Fragrance Ingredient" and "Skin-Conditioning Agents - Emollient" are for topical products only Reference: ICID



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REGULATORY STATUS OF THE INGREDIENT FOR BORDER-LINE APPLICATIONS NATURAL HEALTH PRODUCTS – CANADA [NHPID](#)

Status	NHPID name	Route of administration/ Restrictions
Under the Natural Health Products Regulations, which came into effect on January 1, 2004, natural health products (NHPs) are defined as: Probiotics; Herbal remedies; Vitamins and minerals; Homeopathic medicines; Traditional medicines such as traditional Chinese medicines; Other products like amino acids and essential fatty acids. The Natural Health Products Ingredients Database (NHPID) lists acceptable medicinal and non-medicinal ingredients used in Natural Health Products.		

8. SAFETY ASSESSMENT

CIR COMPENDIUM (Cosmetic Ingredient Review) (http://www.cir-safety.org)			
Denomination	Reference	Maximum "as used" concentration	Conclusions
Glyceryl oleate	IJT 23 (Suppl. 2:55-94), 2004	Up to 5%	Based on the available animal and clinical data included in the report, the CIR Expert Panel concludes that the following Glyceryl Monoesters (Glyceryl linoleate ...) are safe as cosmetic ingredients in the present practices of use and concentrations.

9. CHEMICAL REGULATIONS

CHEMICAL REGULATION			
INCI NAME	CAS #	EINECS #	Chemical name
Glyceryl Oleate	68424-61-3 (or 25496-72-4) (or 91744-20-6) (or 37220-82-9)	270-312-1 (or 247-038-6) (or 294-582-5) (or 253-407-2)	Glycerides, C16-18 and C18-unsatd. mono- and di- (or 9-Octadecenoic acid (9Z)-, monoester with 1,2,3-propanetriol) (or Glycerides, C16-18 and C18-unsatd. mono-, di- and tri-) (or 9-Octadecenoic acid (9Z)-, ester with 1,2,3-propanetriol)

NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDSL)	Australia (AIIC)	China (IECSC)	Other inventories
Glyceryl Oleate	Listed	DSL listed	Listed	Listed	Glycerides, C16-18 and C18-unsatd. mono- and di- : DSL, EINECS, NZIoC, PICCS, REACH, TCSI, TSCA, IECSC, AIIC, AREC, ECL, PICCS, TDCA, VNECI) (or 9-Octadecenoic acid (9Z)-, monoester with 1,2,3-propanetriol (INSQ, NZIoC, PICCS, SWISS, TCSI, TSCA, DSL, ENCS, IECSC, EINECS, AIIC, AREC, ECL) (or Glycerides, C16-18 and C18-unsatd. mono-, di- and tri- (EINECS, REACH, TDCA, VNECI, AREC, ECL, DSL, NZIoC, TCSI, IECSC) (or 9-Octadecenoic acid (9Z)-, ester with 1,2,3-propanetriol: NZIoC, PICCS, TCSI, TDCA, TSCA, DSL, EINECS, IECSC, AIIC, AREC, ECL, PICCS, VNECI)
U.S.A.: Substances that are regulated under other federal regulations, including food additives, drug, cosmetics... are not subject to TSCA regulation. "Naturally occurring substances" are not subject to Premanufacture Notice reporting.					
Canada: Substances not listed on DSL or NDSL can be imported in Canada in quantities not exceeding 100 kg per 12-month period and per importer. Substances listed on NDSL only can be imported in Canada in quantities not exceeding 1000 kg per 12-month period and per importer. Please contact us if you go beyond these volumes. In addition, substances not listed in DSL or NDSL, but listed in the "in-commerce list" can be freely used in					



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NATIONAL INVENTORIES					
INCI NAME	U.S.A. (TSCA)	Canada (DSL/NDL)	Australia (AIC)	China (IECSC)	Other inventories
Products Regulated Under the Food and Drugs Act (F&DA). In addition, substances considered as "Natural" according Health Canada's definition are exempted from New Substances regulation.					
Australia: The Australian Inventory of Chemical Substances (AICS) has been replaced by the Australian Inventory of Industrial Chemicals (AIIC) since July 2020. Chemicals listed on the AIIC are available for industrial use in Australia; however, importers must register with Australian authorities before importing the chemical in Australia. Some chemicals require specific information to be provided to Australian authorities. Naturally occurring chemicals are exempted from AIIC listing; importers may have to submit a declaration. Chemicals not listed on the AIIC must be assessed before importation.					

REACH REGULATION (REGISTRATION, EVALUATION, AUTHORISATION AND RESTRICTION OF CHEMICALS)		
INCI NAME	Status of the product	Status per substance
Glyceryl Oleate	Exempted	Substances intended to be used in medicinal products (human and/or veterinary use) are exempted from the main provisions on registration, evaluation and authorization, under REACH regulation.

10. HISTORY OF USE

10.1. Pharmaceutical applications

10.1.1. FDA IIG reference for Chemical equivalent

FDA Inactive ingredient guide (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
Glyceryl Oleate (UNII: 4PC054V79P)	ORAL	CAPSULE	NA	
	ORAL	POWDER, FOR SUSPENSION	NA	
	ORAL	TABLET		3 mg
	TOPICAL	CREAM, AUGMENTED	3.5%w/w	
	TOPICAL	GEL	1%w/w	
	TOPICAL	SPRAY	0.95%w/w	
	TRANSDERMAL	FILM, EXTENDED RELEASE	18.8mg	
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				



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10.1.1. FDA IIG references to the most similar chemical (read-across approach)

In the **read-across approach**, endpoint information for one chemical (the source chemical) is used to predict the same endpoint for another chemical (the target chemical), which is considered to be "similar" in some way (usually on the basis of structural similarity or on the basis of the same mode or mechanisms of action). In principle, read-across approach can be used to assess physicochemical properties, toxicity, environmental fate and ecotoxicity.

For any of these endpoints, read-across approach may be performed in a qualitative or quantitative manner.'

The similarities may be based on the following:

- a common functional group (e.g. aldehyde, epoxide, ester, specific metal ion);
- common constituents or chemical classes, similar carbon range numbers;
- an incremental and constant change across the category (e.g. a chain-length category);
- the likelihood of common precursors and/or breakdown products, via physical or biological processes, which result in structurally similar chemicals.

These similarities justify data gap filling in a chemical category by applying read-across.

FDA Inactive ingredient guide (http://www.accessdata.fda.gov/scripts/cder/iig/index.cfm)				
Inactive ingredient	Route	Dosage form	Maximum Potency per unit dose	Maximum Daily Exposure (MDE)
GLYCERYL MONOOLEATE (UNII : C4YAD5F5G6)	ORAL	FILM, SOLUBLE		2 mg
Maximum potency and maximum daily exposure: Level info listed in the FDA IIG database refers to a specific product on the market. These levels do not constitute a regulatory max use level. When there is no info, the maximum potency per unit dose and Maximum Daily Exposure (MDE) fields will be blank.				

10.1.2. Other references

HANDBOOK OF PHARMACEUTICAL EXCIPIENT (Current edition by R. C. Rowe, P.J. Sheskey, W. G. Cook, M. E Fenton)			
Denomination	Functional Category	Regulatory status	Safety
Glyceryl monooleate	Emollient; emulsifying agent; emulsion stabilizing agent; gelling agent; modified-release agent; mucoadhesive; nonionic surfactant.	GRAS listed. Included in the FDA Inactive Ingredients Database (oral capsules, oral powder, oral tablets; creams, controlled-release transdermal films). Included in nonparenteral medicines licensed in the UK. Included in the Canadian Natural Health Products Ingredients Database.	Glyceryl monooleate is used in oral and topical pharmaceutical formulations and is generally regarded as a relatively nonirritant and nontoxic excipient.



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COMPENDIUM OF PHARMACEUTICAL EXCIPIENTS FOR VAGINAL FORMULATIONS		
Denomination	Category (concentration in %)	regulatory status
Glycerides of fatty acids	Suppository bases	1, 3–14, 16–26
Abbreviations: G: GRAS, I: inactive ingredients guide, Pharmacopeias, 1: Austrian, 2: Belgian, 3: British, 3v: British (veterinary), 4: Brazilian, 5: Chinese, 6: former Czechoslovakian, 7: Egyptian, 8: European, 9: French, 10: German, 11: Greek, 12: Hungarian, 13: Indian, 14: Italian, 15: International, 16: Japanese, 17: Mexican, 18: Netherlands, 19: Nordic, 20: Portuguese, 21: Romanian, 22: Russian, 23: Swiss, 24: Turkish, 25: United States, 26: former Yugoslavian Garg et al., <i>Pharmaceutical Technology, Drug Delivery</i>, 2001, p14–24, Compendium Of Pharmaceutical Excipients For Vaginal Formulations		

10.2. Food/Nutraceutical/Dietary supplement applications

Mono- and Diglycerides (including Glyceryl oleate) have a long history of use worldwide as food additives for use in food, nutraceutical products and dietary supplements.

COUNTRY	REFERENCE	REGISTRATION NUMBER
International	Evaluations Performed by Joint FAO/WHO Expert Committee on Food Additives	INS 471 Mono- and Diglycerides
E.U.	EFA (European Food additives)	E471 Mono- and Diglycerides
U.S.A.	FCC (Food Chemical Codex)	Mono- and Diglycerides
U.S.A.	GRAS Generally recognized as safe according to 21 CFR	21CFR §184.1505 Mono- and Diglycerides
JAPAN	JSFA (Japanese Standards of Food Additives)	Glycerol esters of fatty acids

HANDBOOK of FOOD ADDITIVES		
Denomination	Use	Safety
PECEOL Glyceryl oleate	Food additive	USP/NF, EP compliance; FDA 21CFR §184.1505 GRAS, FCC, E471, JSFA approvals

NB: The Food additive details are given for information. Gattefossé does not promote the use of this ingredient for Nutraceutical applications (food additive monograph conformity is not guaranteed). The suitability of the ingredient for use in nutraceutical products and dietary supplements remains the own responsibility of the company marketing the finished product.



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